



RESEARCH AND DEVELOPMENT POLICY

Preamble:

The future of any industry, college and university is dependent on innovation index and its linkages with providing the solutions to common problems in a society. With the advent of technologies online resources are available in abundance to each consumer, which make consumers more oriented towards their desired solutions and mounts the pressure on industry to provide more consumer-oriented solutions. So, it is inevitable to change our role from a typical solutions provider and go much beyond the traditional problem-solving approach and provide personalized solutions. But the personalized solution comes with premium which consumers are reluctant to pay due several available options and competitors. There is a dire need to transform the solution development model.

Purposes

This policy is developed in accordance with the Vision mission and the belief with the purpose to create a vibrant atmosphere to promote research among students, faculty and researchers. The research committee has highlighted the need of the research policy to improve the research profile of the institute.

Scope

The research policy is applicable and mandatory to all faculty, research scholars and students of all the departments of SSGMCE.

VISION:

To nurture the research culture at SSGMCE by promoting research in emerging and challenging areas of Engineering, Technology, Science and Management.

MISSION:

1. To develop and foster the Research Ecosystem at institute
2. To develop a support system for researchers in terms of infrastructure, facilities and research funds
3. To collaborate with renown research organizations, leading industries and academic institutions of repute
4. To encourage and motivate the students and faculties to undertake research projects in their area of interest and improve the quality of academic research
5. To improve the quantity and quality of research publications, products and patents

Implementation Plan

Students are the key partner for any innovation, research and entrepreneurial activities. Often, they are left out of innovation and research. The innovation and research mindset of the students need to be leveraged to build the culture of innovation, research and entrepreneurship at SSGMCE.

There is a need to motivate and engage all students into research activities.

The strategy for involving the students in academic research, innovation and entrepreneurship (IR & E) is as follows.



Shri Gajanan Shikshan Sanstha's
SHRI SANT GAJANAN MAHARAJ COLLEGE OF ENGINEERING
SHEGAON – 444203, DIST. BULDANA (MAHARASHTRA STATE), INDIA

"Recognized by A.I.C.T.E., New Delhi" Affiliated to Sant Gadge Baba Amravati University, Amravati
"Approved by the D.T.E., M.S. Mumbai",

Ph : +918669638081/82

Email: principal@ssgmce.ac.in, registrar@ssgmce.ac.in

Website- www.ssgmce.ac.in

Step 1: Each department to identify the department level and interdisciplinary areas for IR & E. Each focus area should form the Similar Interest Groups (SIG)/Faculty Cluster around the activities to be carried out in that specific area. The SIG should include the members from faculty having research interest/skills in given SIG, Students, Alumni and relevant outside experts. Each SIG should present their activities to all the students in a common event two times every year inviting them to join the SIG. All SIGs should maintain the project definition format as per **Annexure: I, it will help them to enroll the students, monitor progress and give equivalence to academic activities e.g., Assignments, Projects or other academics.**

The successful SIGs should host Awareness and Information sharing sessions for end users, prospective SIG members and other relevant populations. Beside this department to host department level hackathons, project, and business plan competition in line with the collective focus of all department SIGs. (If required SGIRAC can help to identify the department specific challenges/projects).

Time Frame: To be carried out first week of August and First week of February every year.

Step 2: Identified SIGs having potential projects for hackathons, publication, patents, grant application, start-up should be nurtured accordingly. Department level single point of contact from faculty and students should meet once in 15 days to discuss, monitor and manage the activities.

Step 3: Launch Event: Public Exhibition of the IR & E outcomes

Organize an open to public event where all the SIGs will exhibit their IR & E outcomes, it will help them to understand needs of the solution, test the demand in actual market, secure funding/investment, tech transfer and make the end user aware.

SGIRAC can help the potential projects in the following ways.

- i. Establish required collaboration
- ii. Identify the suitable funding/investment options
- iii. Acquiring necessary IPRs and Incubation of the business
- iv. Launch, Tech Transfer, Mentoring and Necessary Training of SIGs
- v. Maturing the project into product or service as applicable.
- vi. Pilot scale implementation and test marketing
- vii. Completing legal and technical formalities for Spin-off
- viii. Spin-off launching

Step 4: Final Year Projects and Internship

Each module of Section A of Annexure 1 can be offered as a project to final year UG/PG students, PhD Scholars or other relevant groups.

The all SIGs host a workshop for all 3rd year students every year to motivate them to take the predefined project of the SIGs, the cross-department teams should be encouraged to promote interdisciplinary approach.

The outcome of assigned module should be published in a reputed relevant journal, conference, technical event or obtain necessary IPR



Shri Gajanan Shikshan Sanstha's
SHRI SANT GAJANAN MAHARAJ COLLEGE OF ENGINEERING
SHEGAON - 444203, DIST. BULDANA (MAHARASHTRA STATE), INDIA

"Recognized by A.I.C.T.E., New Delhi" Affiliated to Sant Gadge Baba Amravati University, Amravati
"Approved by the D.T.E., M.S. Mumbai",

Ph : +918669638081/82

Email: principal@sngmce.ac.in, registrar@sngmce.ac.in

Website- www.sngmce.ac.in

To execute Final year under graduate projects the detailed plan of action and methodology to be followed is mentioned in **Annexure II**

Incentives

The incentives to be offered as per the following criteria

SN	Score as per Annexure III	Incentive
1	Less than 20	No incentives
2	Between 21 to 30	Rs 5000/-
3	Between 31 to 45	Rs 10000/-
4	Between 46 to 60	Rs 15000/-
5	Between 61 to 75	Rs 20000/-
6	Between 76 to 80	Rs 25000/-
4	More than 80	Rs 30000/-

Objectives:

- To inculcate research culture at UG level and strengthen at PG level through credit courses in line with NEP 2020.
- To leverage the Choice Based Credit System to Nurture Innovation
- To Leverage NEP 2020 provisions and create conducive environment across the departments to promote research and provide required support facility through research framework and guidelines.
- To organize seminars/conferences/workshops/training programs on cutting edge research/technology to acquire the necessary skill set/knowledge.
- To publish research/review/reports in high quality peer reviewed indexed journals/books
- To fetch research grants from Govt. /Private Industries and consultancy projects to generate funds.
- To develop and strengthen research facilities within campuses.
- To establish partnership/collaboration with one of the top 100 global universities recognized for its academic and research excellence, Industries and SMEs.
- To develop a support system to help researchers to bring their research to society.
- Emerge as a Centre of excellence in creating knowledge.

Areas covered under policy

- PhD Research
- Joint Research Projects of Faculty and Students
- Publication by faculty and students
- Consultancy



Shri Gajanan Shikshan Sanstha's
**SHRI SANT GAJANAN MAHARAJ COLLEGE OF ENGINEERING
SHEGAON – 444203, DIST. BULDANA (MAHARASHTRA STATE), INDIA**

"Recognized by A.I.C.T.E., New Delhi" Affiliated to Sant Gadge Baba Amravati University, Amravati
"Approved by the D.T.E., M.S. Mumbai",

Ph : +918669638081/82

Email: principal@ssgmce.ac.in, registrar@ssgmce.ac.in

Website- www.ssgmce.ac.in

- Proposal for Funding
- Thrust areas for research
- Procedure for
 - Collaboration
 - Organizing and attending the conference
 - Faculty training for research
 - Management of the research policy
- Non-Faculty Research Appointments
- Establishing Independent Laboratories and Research Centers

Seed Money

- SSGMCE should have provision of research grants under the title "Self-Funded Minor Research Projects" for early-stage researchers. The objective of the grant is to provide financial support for pilot scale validation of the project and motivate the faculty members to take up research on priority basis.
- Further Research Cell shall guide respective faculty for submitting the pilot scale validation to suitable funding agencies for higher level funding.
- To promote and motivate faculty clusters for quality publications, Article Processing Charges/Conference Publication fees will be borne by the institute (Once in a year)

Collaborative research/Industry sponsored /Interdisciplinary research

- Collaborations with reputed national and international universities, research organizations, government organizations, industries, NGOs, SMEs. Etc. The professor of all the departments shall take the initiatives and establish linkages with appropriate organization pertaining to the focus areas of their respective departments, leading to consultancy, collaborative projects, faculty/student exchange and other appropriate activities. This will open new avenues for students and faculties to develop networking with researchers/scientists and industry experts. The collaborative projects and consultancy with industry shall aim at developing the solutions/service for solving the industrial/societal problems. Interdisciplinary research focus is to be appreciated as it will develop multifaceted scientists.

Grant Management and IPR Cell needs to be established,

- To prepare, interprets, negotiates, and accepts agreements for funded projects. Liaising activities for Govt. and sponsored programs
- To evaluate the commercial potential of new technologies, determining patentability, prosecuting patents, registering copyrights, marketing and licensing patents, tangible material, and software, negotiating license agreements, and receiving and distributing royalties and other income to the inventors
- To file and manage IPR
- To manage financial conflict of interest
- Integrity Quality and Ethics
- Data privacy and protection of identity



Shri Gajanan Shikshan Sanstha's
SHRI SANT GAJANAN MAHARAJ COLLEGE OF ENGINEERING
SHEGAON – 444203, DIST. BULDANA (MAHARASHTRA STATE), INDIA

"Recognized by A.I.C.T.E., New Delhi" Affiliated to Sant Gadge Baba Amravati University, Amravati
"Approved by the D.T.E., M.S. Mumbai",

Ph : +918669638081/82

Email: principal@ssgmce.ac.in, registrar@ssgmce.ac.in

Website- www.ssgmce.ac.in

- Develop policy for exchange of material, research trials, publication and result sharing

Non-Faculty Research Appointments

Dedicated Appointment Researchers and Academic Staff - Research (AS-R):

Research Scientist, Professors, Research Engineer, Research Scholar or Professor of Practices (as envisioned in NEP 2020)

One research staff/teaching assistant "through earn and learn scheme" shall be appointed for each PI of potential project before the funding and as required/approved in the funding proposal after the funding. Their primary role is to assist the PI in attaining the goals of the PI's research projects. They may participate in the preparation of proposals, progress, and final reports, teaching activities and may be the co-author or sole author of research results, with agreement of the PI. Although not usually engaged in formal classroom teaching, AS-R frequently assist in the guidance of researchers under respective PI, graduate and undergraduate students in the laboratory.

Thrust Areas for 2022-2027 (Subjected to change as per recommendations of SIGs of SSGMCE)

Thrust areas are identified with the help of inputs received from research supervisors and faculty members of the institute.

Engineering • Big Data Analytics • Artificial Intelligence • Sensor Technology • Smart Technologies • Branch Specific Areas identified by the department	Management • Human Resources Management • Finance Management • E-Commerce • Corporate Social Responsibility • Digital Marketing and Social Media Promotion
Interdisciplinary	
• Mother and Child Health • Water Sanitization and Hygiene • Epidemics and Infectious Disease • Tackling AMR • Vaccines and medicines for the communicable and non-communicable diseases	• ICT in Healthcare • ICT in Management • Wearable Technology for Care Delivery • Disease Surveillance Using Technology • Smart Supply Chain Solutions • Biomedical Engineering



Shri Gajanan Shikshan Sanstha's
SHRI SANT GAJANAN MAHARAJ COLLEGE OF ENGINEERING
SHEGAON – 444203, DIST. BULDANA (MAHARASHTRA STATE), INDIA

"Recognized by A.I.C.T.E., New Delhi" Affiliated to Sant Gadge Baba Amravati University, Amravati
"Approved by the D.T.E., M.S. Mumbai",

Ph : +918669638081/82

Email: principal@sngmce.ac.in, registrar@sngmce.ac.in
Website- www.sngmce.ac.in

RESPONSIBILITIES:

1. To prepare policies, procedures and guidelines relating to research and development activities
2. To prepare R&D budget for the institute that majorly includes publication, training, travel and creating new facility for research.
3. To identify the potential areas of research in various disciplines of engineering, science and management and form the faculty clusters based on their specialization.
4. To help faculties to prepare and submit proposals to various funding agencies like AICTE, UGC, DST etc. to obtain grants for projects (In coordination with Bureau of Proposals)
5. To coordinate the matters related to PhD including administration and evaluation of Ph.D. students
6. To encourage and motivate the faculties to register for PhD
7. To encourage faculties to publish their research work in reputed journals having good impact factor and indexing
8. To encourage and motivate students and faculties to attend/publish papers in various National/International Conferences
9. To arrange/organize the FDP/STTP/workshops related to research and development activities
10. To support faculties and students to enhance their knowledge about Intellectual Property Rights (IPR), Patents and copyrights
11. To encourage departments to undertake more MoUs with leading industries and reputed organizations for upskilling of students and faculties and collaborative research
12. To ensure the effective and optimal use of resources by providing platform for sharing available research facilities
13. Preserve code of ethics while publishing papers / thesis
14. To form the student research teams at department/institute level and enhancing the culture of interdisciplinary research
15. To identify potential project/idea that can be transformed into product/patent
16. To motivate faculty, staff and students for taking up startups as per innovation and entrepreneurship policy.

ROLE OF COORDINATORS:

1. To identify thrust/emerging areas of research in consultation with members of department and form the clusters of faculties with respect to their area of specialization/interest.
2. To prepare departmental Rand D budget.
3. To identify students research coordinator from each class.
4. To encourage and support faculty, staff and students to prepare and submit proposals to funding agencies.



Shri Gajanan Shikshan Sanstha's
**SHRI SANT GAJANAN MAHARAJ COLLEGE OF ENGINEERING
SHEGAON – 444203, DIST. BULDANA (MAHARASHTRA STATE), INDIA**

"Recognized by A.I.C.T.E., New Delhi" Affiliated to Sant Gadge Baba Amravati University, Amravati
"Approved by the D.T.E., M.S. Mumbai",

Ph : +918669638081/82

Email: principal@sngmce.ac.in, registrar@sngmce.ac.in

Website- www.sngmce.ac.in

5. To motivate and mentor faculty, staff and students to publish papers in reputed conferences and journals having good impact factor and indexing
6. To ensure proper execution of PhD related activities (Coursework and Evaluation)
7. To make students aware about the expertise, facilities and research activities of the department.
8. To identify and prepare a list of students eligible for incentives related to research activities as per academic incentive scheme.
9. To verify and submit applications of faculties eligible for incentives for research activities to the Chief Coordinator.
10. To take a monthly review of research activities at the department level.

ROLE OF CHIEF COORDINATOR:

1. To prepare R and D budget of the institute and submit to the Head of Institute/Principal for the approval.
2. To prepare/revise the guidelines and policies for research activities and submit to Head of Institute/Principal for the approval
3. To arrange the STTP/FDP/Workshops related to research activities at institute level.
4. To perform administrative responsibilities related to PhD
5. To organize orientation workshops twice in a year. (February and august, where SIG head will present their ongoing research activity and invite interdisciplinary collaborations)
6. To approve and submit a list of students eligible for incentives related to research activities
7. To approve and submit a list of faculties eligible for incentives related to research activities
8. To take review of research activities at institute level in last week of every month.

IMPORTANT GUIDELINES FOR RESEARCH AND DEVELOPMENT ACTIVITIES

1. All the faculty, staff and students should be the part of faculty cluster of their interest with information to R and D Coordinator.
2. SIG/faculty cluster should decide their research focus in mutual consent of all the members once in a year.
3. Each SIG should work in close coordination and submit at least one funding proposal.
4. PhD supervisor with the assistance of their PhD scholars shall write and submit Research Proposal to various funding agencies for financial grant/support.
5. It is mandatory for the department of Applied Science and Humanities and Business Management to submit one research proposal per year for minimum of 5,00,000 Rs
6. It is mandatory for all the Engineering departments to submit one research proposal per year for minimum of 10,00,000 Rs
7. It is mandatory for all the faculties to publish at least one paper per year in UGC CARE/SCOPUS indexed Journal/Conference



Shri Gajanan Shikshan Sanstha's
SHRI SANT GAJANAN MAHARAJ COLLEGE OF ENGINEERING
SHEGAON – 444203, DIST. BULDANA (MAHARASHTRA STATE), INDIA

"Recognized by A.I.C.T.E., New Delhi" Affiliated to Sant Gadge Baba Amravati University, Amravati
"Approved by the D.T.E., M.S. Mumbai",

Ph : +918669638081/82

Email: principal@ssgmce.ac.in, registrar@ssgmce.ac.in

Website- www.ssgmce.ac.in

8. All R&D coordinators and HODs to motivate and encourage their faculty members to publish research articles in SCI/WOS indexed journals
9. All R&D coordinators and HODs to motivate and encourage their faculty members to publish books/book chapters
10. It is mandatory for all the faculties and ME/PhD scholars of the SSGMCE to maintain plagiarism level 15% or as per the guidelines of respective publishers.
11. It is mandatory for all the faculties to attend at least one research training per year.
12. Member should not be allowed to change SIG as they are expected to spend sufficient time on their area of interest.
13. All the Doctorate faculties shall help and support colleagues to pursue their doctoral research.
14. Faculties conducting research shall guide and utilize the SIG effectively.
15. Student research coordinator shall encourage their classmates to actively participate in the SIG research activities.
16. Inter departmental interaction of faculties and student research teams shall be encourage to enhance multi-disciplinary research/projects
17. PhD supervisors and scholar shall acquire the membership of professional societies like IEEE, IE, ASME ISTE etc.

RESEARCH AND DEVELOPMENT POLICIES

To foster the culture of research amongst the faculty members and students some motivational measures are proposed which herein after referred as **Best Practices for Research and Publications**

BEST PRACTICES FOR RESEARCH AND PUBLICATIONS

A. PUBLICATION IN JOURNAL

1. The papers published in journal indexed by ICI/SCOPUS/SCI(WOS)/SCIE/ SCImago shall only be considered for Article processing charges (APC) reimbursement
2. In case of multiple authors all authors shall be affiliated to SSGMCE/Corresponding Author shall be affiliated to SSGMCE
3. APC for the publication up to Rs 10,000/- for the paper published in SCOPUS/SCI/WoS indexed journal (Once in a year)
4. In no case the publication charges shall be paid by the institute
5. It is mandatory for all faculties of SSGMCE to publish at least one paper per year in a journal of repute



Shri Gajanan Shikshan Sanstha's
SHRI SANT GAJANAN MAHARAJ COLLEGE OF ENGINEERING
SHEGAON – 444203, DIST. BULDANA (MAHARASHTRA STATE), INDIA

"Recognized by A.I.C.T.E., New Delhi" Affiliated to Sant Gadge Baba Amravati University, Amravati
"Approved by the D.T.E., M.S. Mumbai"

Ph : +918669638081/82

Email: principal@ssgmce.ac.in, registrar@ssgmce.ac.in
Website- www.ssgmce.ac.in

B. PUBLICATION IN CONFERENCES IN INDIA

1. The papers published in the proceedings of conferences organized/sponsored by IEEE, IET, SPRINGER, ELSEVIER etc. shall only be considered for reimbursement
2. Faculties of SSGMCE shall be eligible for the sponsorship in the form of registration fees, travel and dearness allowance, and duty leave
3. In case of multiple authors all authors shall be affiliated to SSGMCE and only one author shall be eligible for 100% reimbursement of registration fees for the presentation
4. In case of multiple authors maximum two authors shall get the sponsorship of registration fees (attending only), travel and dearness allowance, and duty leave
5. The sponsorship for the individual faculty/PhD scholar shall be given maximum for one conference paper per year
6. It is mandatory for all faculties of SSGMCE to publish at least one paper per year in either a conference of repute OR publish a book chapter in a book with good indexing

C. PUBLICATION IN CONFERENCES OUTSIDE INDIA

1. Faculties of SSGMCE shall be eligible for the reimbursement for publication/presentation of research paper in a conference outside India
2. Faculties of SSGMCE shall be eligible for the sponsorship of 50% registration fees, duty leave as per norms, travelling allowance (Train/Bus TO and FRO) from Shegaon to city of embarkment (Viz. Mumbai, Delhi, etc.) and one-way airfare from the city of embarkment to conference destination
3. It is mandatory for the faculty applying for sponsorship for presentation abroad to submit a travel grant proposal to AICTE, UGC, SERB etc.
4. In case of multiple authors only one author shall be eligible for the sponsorship
5. Sponsorship for presentation in abroad conference shall be applicable to only that paper where the affiliation of authors is SSGMCE
6. An individual faculty from SSGMCE can apply for the sponsorship for presentation of paper in abroad once in three years
7. Faculty applying for sponsorship have to submit detailed travel plan and necessary documents to Honorable Principal for sanction at least one month in advance

D. PUBLICATION OF PATENT

1. Faculties of SSGMCE shall be eligible for the reimbursement for the publication fees of patents
2. Reimbursement is applicable to only those patents which are filed jointly with SSGMCE
3. No SSGMCE faculty /student can apply for patent personally /individually if the project is sponsored by SSGMCE/SGIARC even after leaving SSGMCE.



Shri Gajanan Shikshan Sanstha's
SHRI SANT GAJANAN MAHARAJ COLLEGE OF ENGINEERING
SHEGAON – 444203, DIST. BULDANA (MAHARASHTRA STATE), INDIA

"Recognized by A.I.C.T.E., New Delhi" Affiliated to Sant Gadge Baba Amravati University, Amravati
"Approved by the D.T.E., M.S. Mumbai",

Ph : +918669638081/82

Email: principal@ssgmce.ac.in, registrar@ssgmce.ac.in

Website- www.ssgmce.ac.in

4. All patents shall be filed through IP cell of SGIARC/SSGMCE wherein the first applicant and hence the owner is SSGMCE. However, the co-applicant may be inventors
5. All the commercial rights of the patent shall be with SGIARC/SSGMCE
6. The fess for filling patent shall be paid by SSGMCE
7. The revenue generated after commercialization of the patent shall be shared between SSGMCE/SGIARC and the inventors however the percentage of sharing will be mutually decided by the SSGMCE/SGIARC and the inventor as per the policies

INCENTIVES FOR SPONSORED RESEARCH, CONSULTANCY

A. SPONSORED RESEARCH

1. It is mandatory for all the departments to submit a research proposal to the funding agencies (AICTE, UGC, DST etc.) every year
2. A research proposal accepted by the funding agencies (AICTE/UGC/DST etc.) shall be regarded as sponsored research
3. Faculties of SSGMCE are eligible for the incentives for sponsored research
4. Incentive of 1% of the total fund sanctioned OR Rs 10000/- (whichever is minimum) shall be applicable to principal investigator
5. In case of multiple applicant/investigators the incentive shall be shared between PI and Co-PI
6. Incentives shall be applicable only when all the formalities/documentation with funding agency are completed in all respect and due time

B. CONSULTANCY POLICY

1. Faculty, staff and student of SSGMCE are eligible for the revenue sharing for providing consultancy services to external world/ industry
2. SSGMCE reserves the right to depute any faculty for consultancy work if the industry approached to Institute
3. Individual member shall obtain prior permission of the Principal before any kind of commitment with industry
4. Completion of consultancy task to the satisfaction of respective agency is responsibility of the members involved.
5. In order to promote industrial consultancy activities, following revenue sharing shall be applicable



Shri Gajanan Shikshan Sanstha's
**SHRI SANT GAJANAN MAHARAJ COLLEGE OF ENGINEERING
SHEGAON – 444203, DIST. BULDANA (MAHARASHTRA STATE), INDIA**

"Recognized by A.I.C.T.E., New Delhi" Affiliated to Sant Gadge Baba Amravati University, Amravati
"Approved by the D.T.E., M.S. Mumbai",

Ph : +918669638081/82

Email: principal@ssgmce.ac.in, registrar@ssgmce.ac.in

Website- www.ssgmce.ac.in

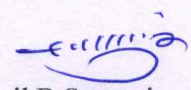
Sr No	Consultancy Activity	Institute Share	Faculty Share
01	Routine Testing and Measurement/Services/Counselling	50%	50%
02	Consultancy received by college and faculty is deputed for the work	40%	60%
03	Consultancy received by faculty and if college facilities not utilized	60%	40%
04	Consultancy received by faculty and if college facilities are utilized	70%	30%

- Subjected to revision in annual review meeting.

RECOGNITION OF FACULTY RESEARCHER

1. In order to recognize the research contribution of the faculty members "**Faculty Researcher of the Year Award**" shall be presented to deserving faculty member
2. To promote and motivate research culture an honor list of faculty members having research acumen shall be prepared and published every year


Dr Saurabh S Jadhao
Chief Coordinator R&D Cell
SSGMCE Shegaon


Dr Sunil B Somani
Principal
SSGMCE Shegaon

PRINCIPAL
Shri Sant Gajanan Maharaj
College of Engineering, Shegaon.

